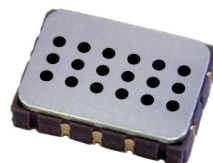




The MiCS-6814 is a compact MOS sensor with three fully independent sensing elements on one package.

The MiCS-6814 is a robust MEMS sensor for the detection of pollution from automobile exhausts and for agricultural/industrial odors.

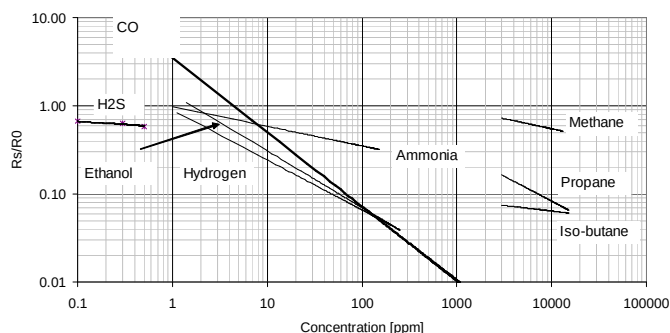


Features

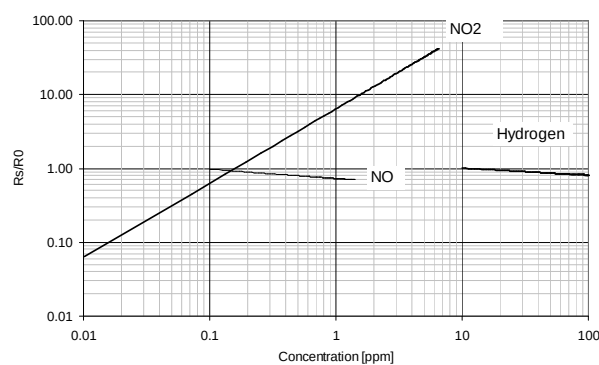
- Smallest footprint for compact designs (5 x 7 x 1.55 mm)
- Robust MEMS sensor for harsh environments
- High-volume manufacturing for low-cost applications
- Short lead-times

Detectable gases

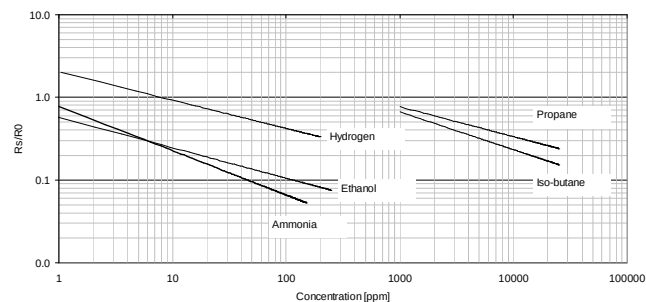
- | | | |
|--------------------|----------------------------------|--------------|
| • Carbon monoxide | CO | 1 – 1000ppm |
| • Nitrogen dioxide | NO ₂ | 0.05 – 10ppm |
| • Ethanol | C ₂ H ₅ OH | 10 – 500ppm |
| • Hydrogen | H ₂ | 1 – 1000ppm |
| • Ammonia | NH ₃ | 1 – 500ppm |
| • Methane | CH ₄ | >1000ppm |
| • Propane | C ₃ H ₈ | >1000ppm |
| • Iso-butane | C ₄ H ₁₀ | >1000ppm |



RED sensor, continuous power ON, 25°C, 50% RH



OX sensor, continuous power ON, 25°C, 50% RH



NH3 sensor, continuous power ON, 25°C, 50% RH

Performance RED sensor

Characteristic RED sensor	Symbol	Typ	Min	Max	Unit
Sensing resistance in air (see note 1)	R_0	-	100	1500	k Ω
Typical CO detection range	FS		1	1000	ppm
Sensitivity factor (see note 2)	S_{60}	-	1.2	50	-

Performance OX sensor

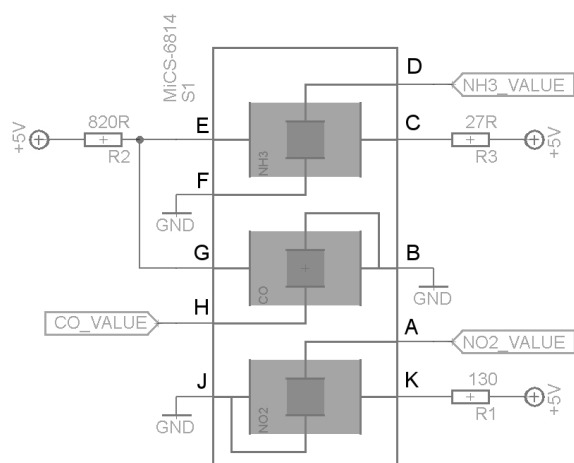
Characteristic OX sensor	Symbol	Typ	Min	Max	Unit
Sensing resistance in air (see note 1)	R_0	-	0.8	20	k Ω
Typical NO ₂ detection range	FS		0.05	10	ppm
Sensitivity factor (see note 3)	S_R	-	2	-	-

Performance NH3 sensor

Characteristic OX sensor	Symbol	Typ	Min	Max	Unit
Sensing resistance in air (see note 1)	R_0	-	10	1500	k Ω
Typical NH ₃ detection range	FS		1	300	ppm
Sensitivity factor (see note 4)	S_R	-	1.5	15	-

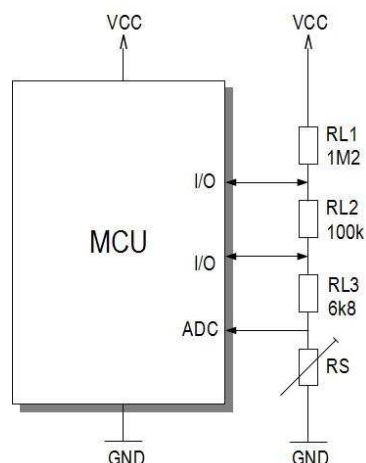
Notes:

1. Sensing resistance in air R_0 is measured under controlled ambient conditions, i.e. synthetic air at 23 $\pm$ 5°C and 50 $\pm$ 10% RH for RED sensor and synthetic air at 23 $\pm$ 5°C and $\leq$ 5% RH for OX sensor. Sampling test.
2. Sensitivity factor is defined as R_s in air divided by R_s at 60 ppm CO. Test conditions are 23 $\pm$ 5°C and 50 $\pm$ 10% RH. Indicative values only. Sampling test.
3. Sensitivity factor is defined as R_s at 0.25 ppm NO₂, divided by R_s in air. Test conditions are 23 $\pm$ 5°C and $\leq$ 5% RH. Indicative values only. Sampling test.
4. Sensitivity factor is defined as R_s in air divided by R_s at 1 ppm of NH₃. Test conditions are 23 $\pm$ 5°C and 50 $\pm$ 10% RH. Indicative values only. Sampling test.



MiCS-6814 with recommended supply circuit (top view)

R1 is 130 Ohms, R2 is 820 Ohms and R3 is 27 Ohms. These resistors are necessary to obtain the right temperatures on the three independent heaters while using a single 5 V power supply. The resulting voltages are typically $V_{hRED} = 2.4$ V, $V_{hOX} = 1.7$ V and $V_{hNH3} = 2.2$ V.

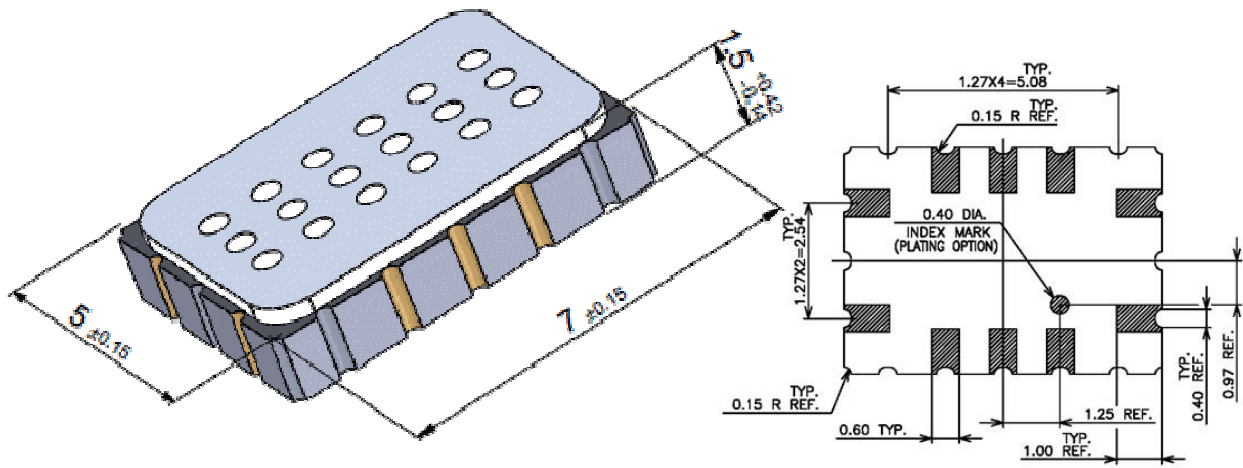


MiCS-6814 with measurement circuit (top view)

The example above shows the principle to read a resistance between 1 kOhms and 2MOhms with a 8bit A/D converter. Depending on the application and on the precision of the A/D converter, a single load resistor of 56 kOhms may be sufficient. Rload must be 820 Ohms at the lowest in order not to damage the sensitive layer.

Parameter RED sensor/OX sensor/NH3 sensor	Symbol	Typ	Min	Max	Unit
Heating power	P_H	76/43/66	71/30/60	81/50/73	mW
Heating voltage	V_H	2.4/1.7/2.2	-	-	V
Heating current	I_H	32/26/30	-	-	mA
Heating resistance at nominal power	R_H	74/66/72	66/59/64	82/73/80	Ω

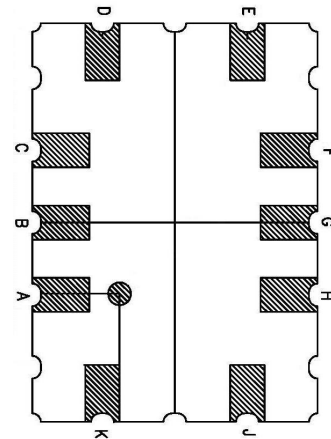
Rating	Symbol	Value / Range	Unit
Maximum heater power dissipation	P_H	88	mW
Maximum sensitive layer power dissipation	P_s	8	mW
Voltage supplyHeating current	V_{supply}	4.9 – 5.1	V
Relative humidity range	RH	5 – 95	%RH
Ambient operating temperature	T_{amb}	-30 – 85	$^{\circ}\text{C}$
Storage temperature range	T_{sto}	-40 – 120	$^{\circ}\text{C}$
Storage humidity range	RHsto	5 - 95	%RH



Package outline dimensions

The package is compatible with SMD assembly process.

Pin	Connection
A	Rs2
B	Rh1 RED, Rs1 RED
C	Rh1 NH ₃
D	Rs1 NH ₃
E	Rh2 NH ₃
F	Rs2 NH ₃
G	Rh2 RED
H	Rs2 RED
J	Rh1 OX, Rs1 OX
K	Rh2 OX

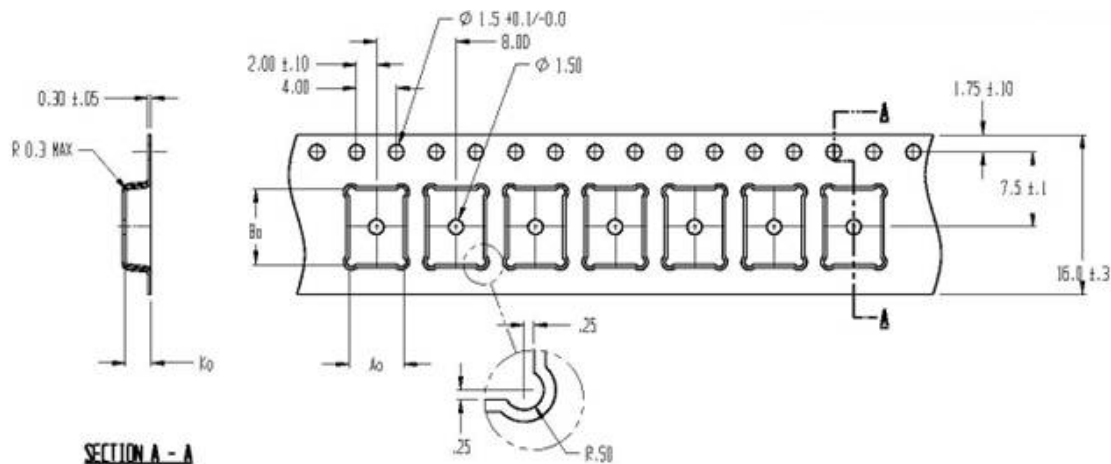


MiCS-6814 configuration (bottom view)

Sensor configuration

The silicon gas sensor structure consists of an accurately micro machined diaphragm with an embedded heating resistor and the sensing layer on top.

The MiCS-6814 includes three sensor chips with independent heaters and sensitive layers. One sensor chip detects oxidising gases (OX), the other sensor detects reducing gases (RED) and the other detects NH₃. The internal connections are shown above.

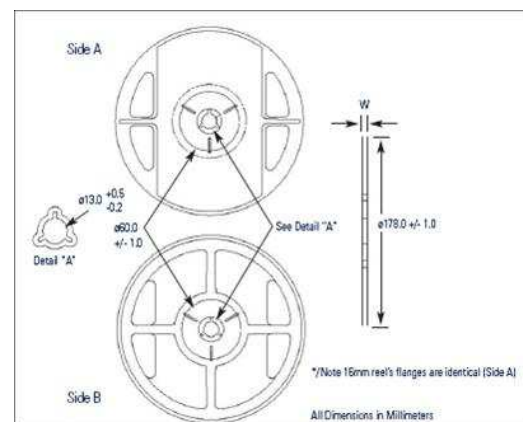


Packaging

The sensors are packaged in a tape and reel for expedition.

The sensors are placed in a carrier type. The dimensions of the cavity are 5.5 x 7.5 x 2.55 mm (the tolerance is +/- 0.2 mm).

The outside dimension of the reel is either 178 +/- mm (for a maximum of 700 sensors) or 330 + 0.25 / -4 mm (for a maximum of 2000 sensors).



IMPORTANT PRECAUTIONS:

Read the following instructions carefully before using the MiCS-6814 described here to avoid erroneous readings and to prevent the device from permanent damage.

- The sensor must be reflow soldered in a neutral atmosphere, without soldering flux vapours.
- The sensor must not be exposed to high concentrations of organic solvents, silicone vapours or cigarette-smoke in order to avoid poisoning the sensitive layer.
- Heater voltage above the specified maximum rating will destroy the sensor due to overheating.
- This sensor is to be placed in a filtered package that protects it against water and dust projections.
- SGX sensortech strongly recommends using ESD protection equipment to handle the sensor.

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